CAPSTONE PROJECT - MARKETING

PROJECT TASK - WEEK 1

1. Dataset Structure
    - Rows (entries) - 9551
    - Columns(features) - 19

    These columns include details like restaurant name, location (city, address, coordinates), cuisines, cost, rating, booking options, and more.

    Column names

```
[1] import pandas as pd

df = pd.read_excel("data.xlsx")
print(df.head(10))

Show hidden output

[ ] print(df.columns)

Index(['Restaurant ID', 'Restaurant Name', 'Country Code', 'City', 'Address',
       'Locality', 'Locality Verbose', 'Longitude', 'Latitude', 'Cuisines',
       'Average Cost for two', 'Currency', 'Has Table booking',
       'Has Online delivery', 'Price range', 'Aggregate rating',
       'Rating color', 'Rating text', 'Votes'],
      dtype='object')
```

Column Datatypes

```
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9551 entries, 0 to 9550
Data columns (total 19 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Restaurant ID         9551 non-null   int64
 1   Restaurant Name       9550 non-null   object
 2   Country Code          9551 non-null   int64
 3   City                  9551 non-null   object
 4   Address               9551 non-null   object
 5   Locality              9551 non-null   object
 6   Locality Verbose      9551 non-null   object
 7   Longitude             9551 non-null   float64
 8   Latitude              9551 non-null   float64
 9   Cuisines              9542 non-null   object
 10  Average Cost for two  9551 non-null   int64
 11  Currency              9551 non-null   object
 12  Has Table booking     9551 non-null   object
 13  Has Online delivery   9551 non-null   object
 14  Price range           9551 non-null   int64
 15  Aggregate rating      9551 non-null   float64
 16  Rating color          9551 non-null   object
 17  Rating text           9551 non-null   object
 18  Votes                 9551 non-null   int64
dtypes: float64(3), int64(5), object(11)
memory usage: 1.4+ MB
None
```

2. Missing Values - Found the duplicates and missing values and dropped the rows with missing values

```
df.isnull().sum()
```

	0
Restaurant ID	0
Restaurant Name	1
Country Code	0
City	0
Address	0
Locality	0
Locality Verbose	0
Longitude	0
Latitude	0
Cuisines	9
Average Cost for two	0
Currency	0
Has Table booking	0
Has Online delivery	0

Handling the missing value

Here we have dropped the row with missing Restaurant name.

```
df.dropna(subset = ["Restaurant Name"])
```

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longitude	La
0	7402935	Skye	94	Jakarta	Menara BCA, Lantai 56, Jl. MH. Thamrin, Thamri...	Grand Indonesia Mall, Thamrin	Grand Indonesia Mall, Thamrin, Jakarta	106.821999	-6.
1	7410290	Satoo - Hotel Shangri-La	94	Jakarta	Hotel Shangri-La, Jl. Jend. Sudirman	Hotel Shangri-La, Sudirman	Hotel Shangri-La, Sudirman, Jakarta	106.818961	-6.
2	7420899	Sushi Masa	94	Jakarta	Jl. Tuna Raya No. 5, Penjaringan	Penjaringan	Penjaringan, Jakarta	106.800144	-6.
3	7421967	3 Wise Monkeys	94	Jakarta	Jl. Suryo No. 26, Senopati, Jakarta	Senopati	Senopati, Jakarta	106.813400	-6.
4	7422489	Avec Moi Restaurant and Bar	94	Jakarta	Gedung PIC, Jl. Teluk Betung 43, Thamrin, Jakarta	Thamrin	Thamrin, Jakarta	106.821023	-6.
...	...	...	...	...	...	...	...	...	...

Here we have dropped rows with missing Cuisine Values.

```
df.dropna(subset = ["Cuisines"])
```

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose	Longitude	La
0	7402935	Skye	94	Jakarta	Menara BCA, Lantai 56, Jl. MH. Thamrin, Thamri...	Grand Indonesia Mall, Thamrin	Grand Indonesia Mall, Thamrin, Jakarta	106.821999	-6.
1	7410290	Satoo - Hotel Shangri-La	94	Jakarta	Hotel Shangri-La, Jl. Jend. Sudirman	Hotel Shangri-La, Sudirman	Hotel Shangri-La, Sudirman, Jakarta	106.818961	-6.
2	7420899	Sushi Masa	94	Jakarta	Jl. Tuna Raya No. 5, Penjaringan	Penjaringan	Penjaringan, Jakarta	106.800144	-6.
3	7421967	3 Wise Monkeys	94	Jakarta	Jl. Suryo No. 26, Senopati, Jakarta	Senopati	Senopati, Jakarta	106.813400	-6.
4	7422489	Avec Moi Restaurant and Bar	94	Jakarta	Gedung PIC, Jl. Teluk Betung 43, Thamrin, Jakarta	Thamrin	Thamrin, Jakarta	106.821023	-6.

3. The geographical distribution of restaurants- The top 10 cities with most restaurants

```
# Top 10 cities with most restaurants
df["City"].value_counts().head(10)
```

City	count
New Delhi	5473
Gurgaon	1118
Noida	1080
Faridabad	251
Ghaziabad	25
Ahmedabad	21
Amritsar	21
Bhubaneshwar	21
Guwahati	21
Lucknow	21

dtype: int64

City with most number of restaurants - Delhi - 5473

City with the least number of restaurant - Bandung - 1

```
# City with least resataurants
df["City"].value_counts(ascending=True)
```

City	count
Bandung	1
Panchkula	1
Mohali	1
Quezon City	1
Tagaytay City	1
...	...
Ghaziabad	25
Faridabad	251
Noida	1080
Gurgaon	1118
New Delhi	5473

141 rows × 1 columns

4. The restaurants with most franchise are:

```
# Restaurants with many franchise
df.groupby("Restaurant Name")["City"].nunique().sort_values(ascending = False)
```

Restaurant Name	City
Barbeque Nation	22
Pizza Hut	12
KFC	8
The Yellow Chilli	7
Domino's Pizza	7
...	...
Fu.D	1
Fry Centre	1
Fruitpro	1
Frugurpop- ibis New Delhi	1
Funk House Cafe	1

5. The ratio of restaurant with table booking vs with no table booking is

```
[29] # Ratio of restaurants having table booking vs no table booking
     ratio = df["Has Table booking"].value_counts().get("Yes", 0) / df["Has Table booking"].value_counts().ge
     print("Table booking ratio(Yes/No):", ratio)

Table booking ratio(Yes/No): 0.13797211962349576
```

6. The percentage of restaurants providing online delivery is 25.6%

```
[18] # Percentage of restaurants providing online delivery
     percentage = (df['Has Online delivery'].value_counts().get('Yes', 0) / df['Has Online delivery'].notna().sum()) * 100
     print(f"{percentage:.2f}% of restaurants provide online delivery.")

25.66% of restaurants provide online delivery.
```

7. The difference between number of votes for the restaurants that deliver and the restaurants that do not deliver is -462817

```
# Difference between number of votes between restaurants which delivers and does not deliver
vote_difference = df[df['Has Online delivery'] == 'Yes']['Votes'].sum() - df[df['Has Online delivery'] == 'No']['Votes'].sum()
print(vote_difference)

-462817
```

PROJECT TASK - Q WEEK 2

1. The top 10 cuisines served across cities :

```python
# Top 10 cuisines served across cities
top_10_cuisines = (df['Cuisines'].dropna().str.split(',').explode().str.strip().value_counts().head(10))
print(top_10_cuisines)
```

```
Cuisines
North Indian    3960
Chinese         2735
Fast Food       1986
Mughlai          995
Italian          764
Bakery           745
Continental      736
Cafe             703
Desserts         653
South Indian     636
Name: count, dtype: int64
```

2. The maximum and minimum number of cuisines that a restaurant serves

```python
# Drop missing cuisines
df_clean = df.dropna(subset=['Cuisines'])

# Count number of cuisines per restaurant
df_clean['Cuisine Count'] = df_clean['Cuisines'].str.split(',').apply(len)

# Get max and min
max_cuisines = df_clean['Cuisine Count'].max()
min_cuisines = df_clean['Cuisine Count'].min()

print("Maximum cuisines served by one restaurant:", max_cuisines)
print("Minimum cuisines served by one restaurant:", min_cuisines)
```

```
Maximum cuisines served by one restaurant: 8
Minimum cuisines served by one restaurant: 1
```

The most served cuisine across the restaurant for each city

```
df_exploded = df_clean.copy()
df_exploded['Cuisines'] = df_exploded['Cuisines'].str.split(',')
df_exploded = df_exploded.explode('Cuisines')
df_exploded['Cuisines'] = df_exploded['Cuisines'].str.strip()

top_cuisine_per_city = (
    df_exploded.groupby(['City', 'Cuisines'])
    .size()
    .reset_index(name='Count')
    .sort_values(['City', 'Count'], ascending=[True, False])
    .drop_duplicates('City')
)

print(top_cuisine_per_city)
```

```
                 City      Cuisines  Count
11           Abu Dhabi        Indian      7
37                Agra  North Indian     15
48           Ahmedabad   Continental     12
62              Albany      American      4
91           Allahabad  North Indian     12
...                ...           ...    ...
1815           Weirton        Burger      1
1820  Wellington City          Cafe      9
1835   Winchester Bay        Burger      1
1838           Yorkton         Asian      1
1842         €¡stanbul          Cafe      4
```

3. The distribution cost across the restaurants

```
bins = [0, 200, 400, 600, 800, 1000, 2000, df['Average Cost for two'].max()]
labels = ['<200', '200-400', '400-600', '600-800', '800-1000', '1000-2000', '2000+']
df['Cost Range'] = pd.cut(df['Average Cost for two'], bins=bins, labels=labels)

cost_distribution = df['Cost Range'].value_counts().sort_index()
print(cost_distribution)
```

```
Cost Range
<200         2117
200-400      2701
400-600      2077
600-800       996
800-1000      467
1000-2000     917
2000+         258
Name: count, dtype: int64
```

4. Various factors affecting rating

```
# Ratings by Online delivery
df.groupby('Has Online delivery')['Aggregate rating'].mean()
```

Has Online delivery	Aggregate rating
No	2.465296
Yes	3.248837

dtype: float64

```
[32] # Ratings by Table booking
df.groupby('Has Table booking')['Aggregate rating'].mean()
```

Has Table booking	Aggregate rating
No	2.559359
Yes	3.441969

dtype: float64

```
# Ratings by City
df.groupby('City')['Aggregate rating'].mean().sort_values(ascending=False)
```

City	Aggregate rating
Inner City	4.900000
Quezon City	4.800000
Makati City	4.650000
Pasig City	4.633333
Mandaluyong City	4.625000
...	...
New Delhi	2.438845
Montville	2.400000
Mc Millan	2.400000
Noida	2.036204
Faridabad	1.866932

141 rows × 1 columns

```
[34] # Ratings by Cuisines (Top 10 Cuisines)
     df_cuisine = df.dropna(subset=['Cuisines'])
     df_cuisine = df_cuisine.copy()
     df_cuisine['Cuisines'] = df_cuisine['Cuisines'].str.split(',')
     df_cuisine = df_cuisine.explode('Cuisines')
     df_cuisine['Cuisines'] = df_cuisine['Cuisines'].str.strip()
     df_cuisine.groupby('Cuisines')['Aggregate rating'].mean().sort_values(ascending=False).head(10)
```

Cuisines	Aggregate rating
Sunda	4.900000
Bİ_rek	4.700000
Taiwanese	4.650000
Ramen	4.500000
Dim Sum	4.466667
Hawaiian	4.412500
Bubble Tea	4.400000
Dİ_ner	4.400000
Curry	4.400000

```
# Ratings by Cost range
bins = [0, 200, 400, 600, 800, 1000, 2000, df['Average Cost for two'].max()]
labels = ['<200', '200-400', '400-600', '600-800', '800-1000', '1000-2000', '2000+']
df['Cost Range'] = pd.cut(df['Average Cost for two'], bins=bins, labels=labels)

df.groupby('Cost Range')['Aggregate rating'].mean().sort_index()
```

```
/tmp/ipython-input-35-1847908449.py:6: FutureWarning: The default of observed=False is deprecated and will be changed to True in a future
  df.groupby('Cost Range')['Aggregate rating'].mean().sort_index()
```

Cost Range	Aggregate rating
<200	2.431790
200-400	2.117401
400-600	2.664420
600-800	3.139859
800-1000	3.337045
1000-2000	3.664231
2000+	3.689147

dtype: float64

5. The overall ratings of restaurants are influenced by several key factors observed in the dataset. One significant factor is the **number of cuisines** a restaurant offers. A broader variety of cuisines can appeal to a wider customer base and enhance customer satisfaction. However, it may also reduce the specialization in any one cuisine, potentially affecting the quality and, consequently, the ratings.

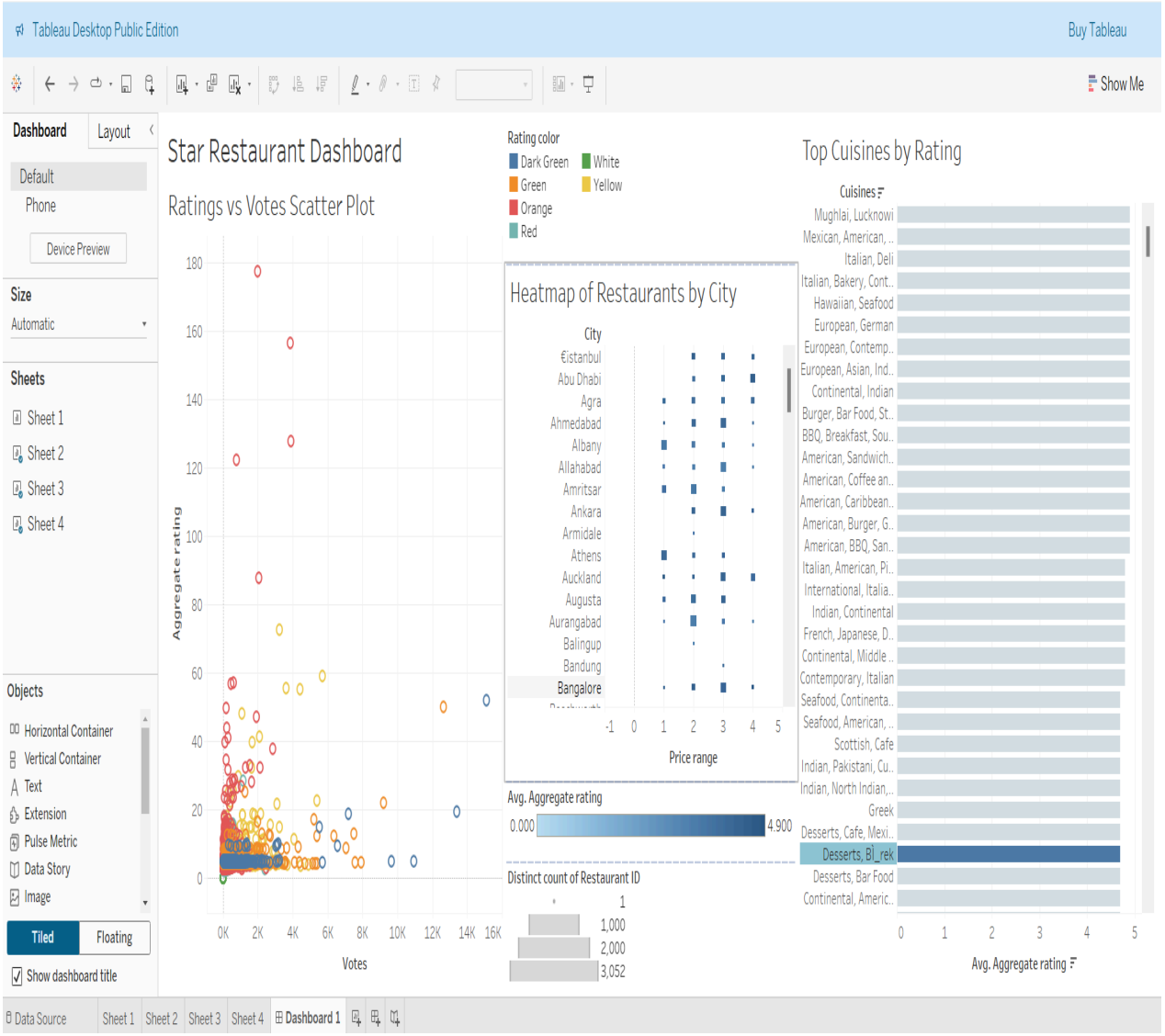
   Another major factor is the **average cost for two people**. Pricing reflects the positioning of the restaurant — low-cost outlets may be popular for affordability but may not offer the same ambiance or quality as mid-range or premium restaurants. Often, restaurants that maintain a balance between affordability and quality tend to receive higher ratings.

   The availability of **online delivery** also plays a vital role. Restaurants that offer delivery services cater to the growing demand for convenience, which can positively impact their ratings, especially among younger and urban customers. Similarly, the **table booking option** indicates a more formal or premium experience. Such restaurants often focus more on service quality and customer satisfaction, which can contribute to higher ratings.

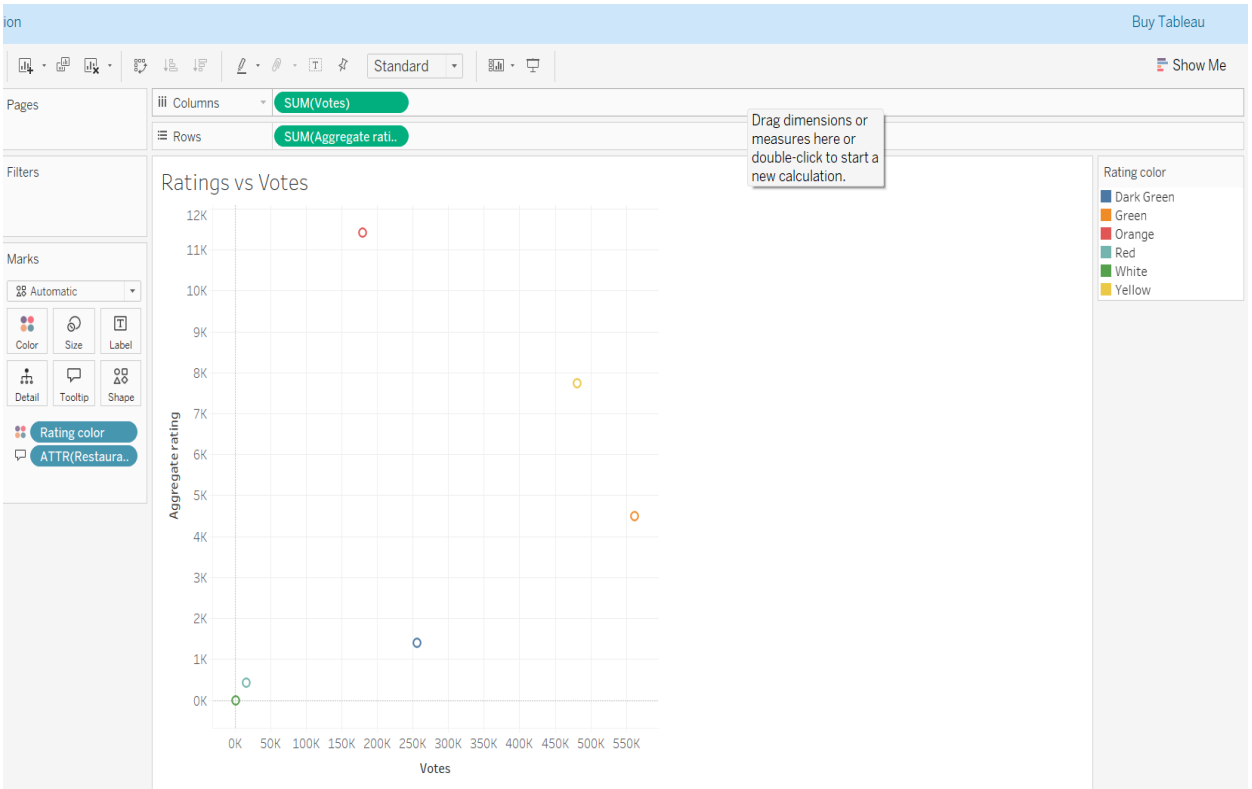
   The **number of votes** received by a restaurant is a direct indicator of customer engagement. Ratings supported by a large number of votes tend to be more reliable, as they represent the collective experience of many diners. Lastly, the **location or city** of the restaurant is an influential factor. Ratings can vary across cities due to differences in customer expectations, cultural preferences, and competition.

   In summary, a restaurant's rating is a reflection of not just its food, but also its pricing, variety, service options, customer base, and local environment. Understanding these factors can help businesses identify opportunities for improvement and position themselves more effectively in the market.

6. Star Restaurant Dashboard



7. Variables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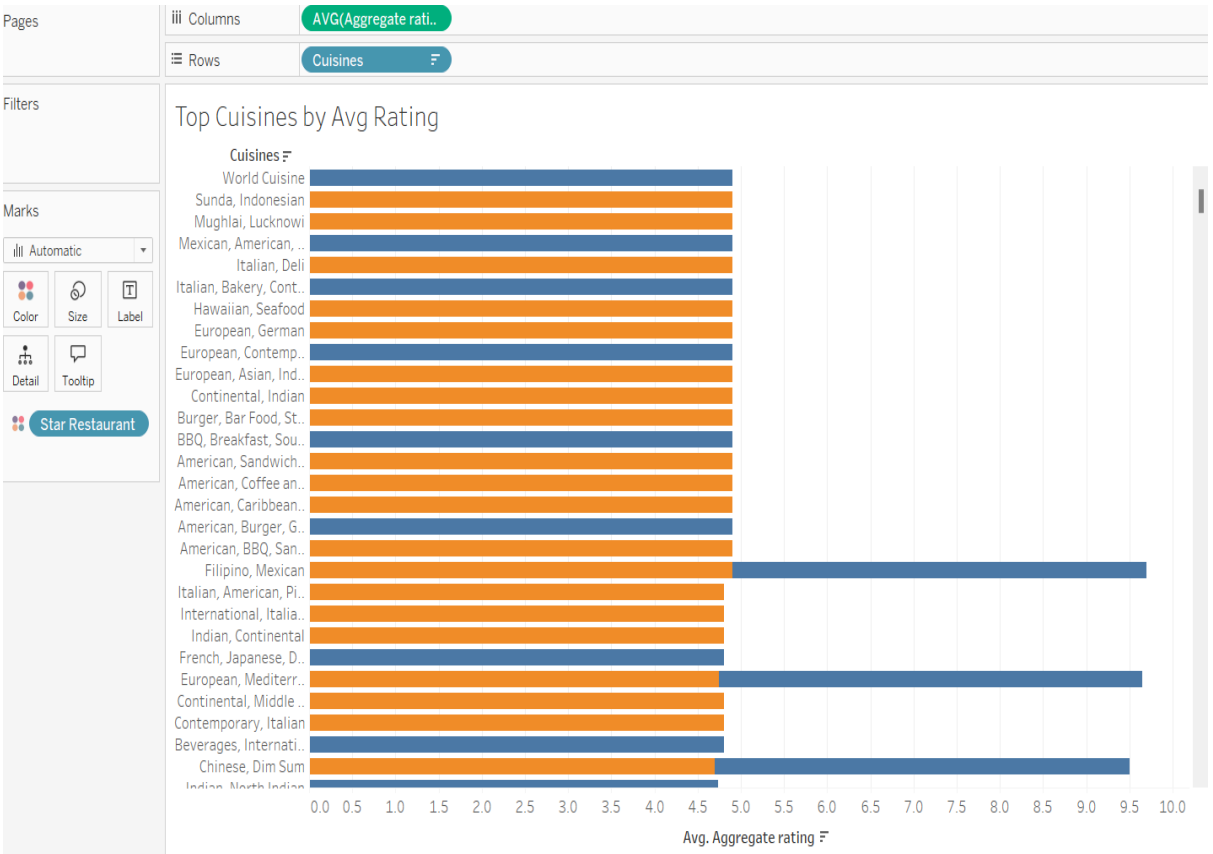
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Star Restaurant
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Cuisines
World Cuisine
Sunda, Indonesian
Mughlai, Lucknowi
Mexican, American, ..
Italian, Deli
Italian, Bakery, Cont..
Hawaiian, Seafood
European, German
European, Contemp..
European, Asian, Ind..
Continental, Indian
Burger, Bar Food, St..
BBQ, Breakfast, Sou..
American, Sandwich..
American, Coffee an..
American, Caribbean..
American, Burger, G..
American, BBQ, San..
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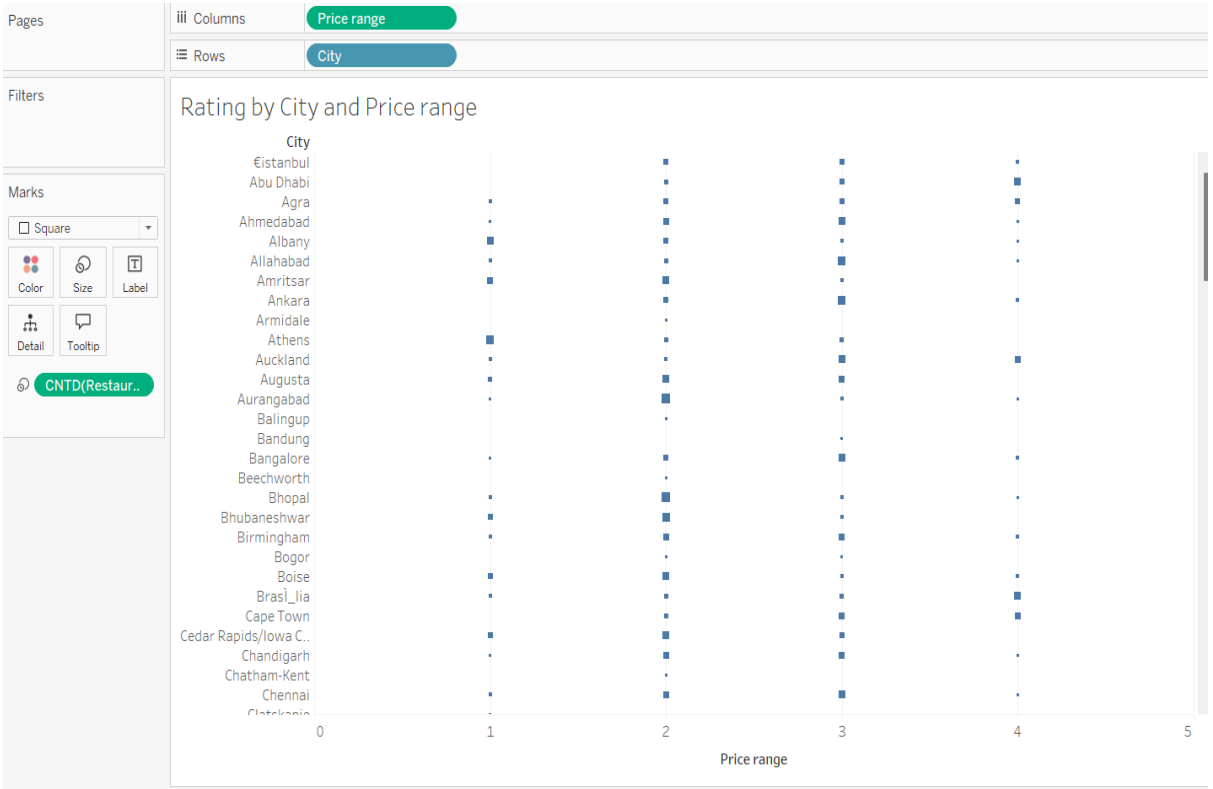
Avg. Aggregate rating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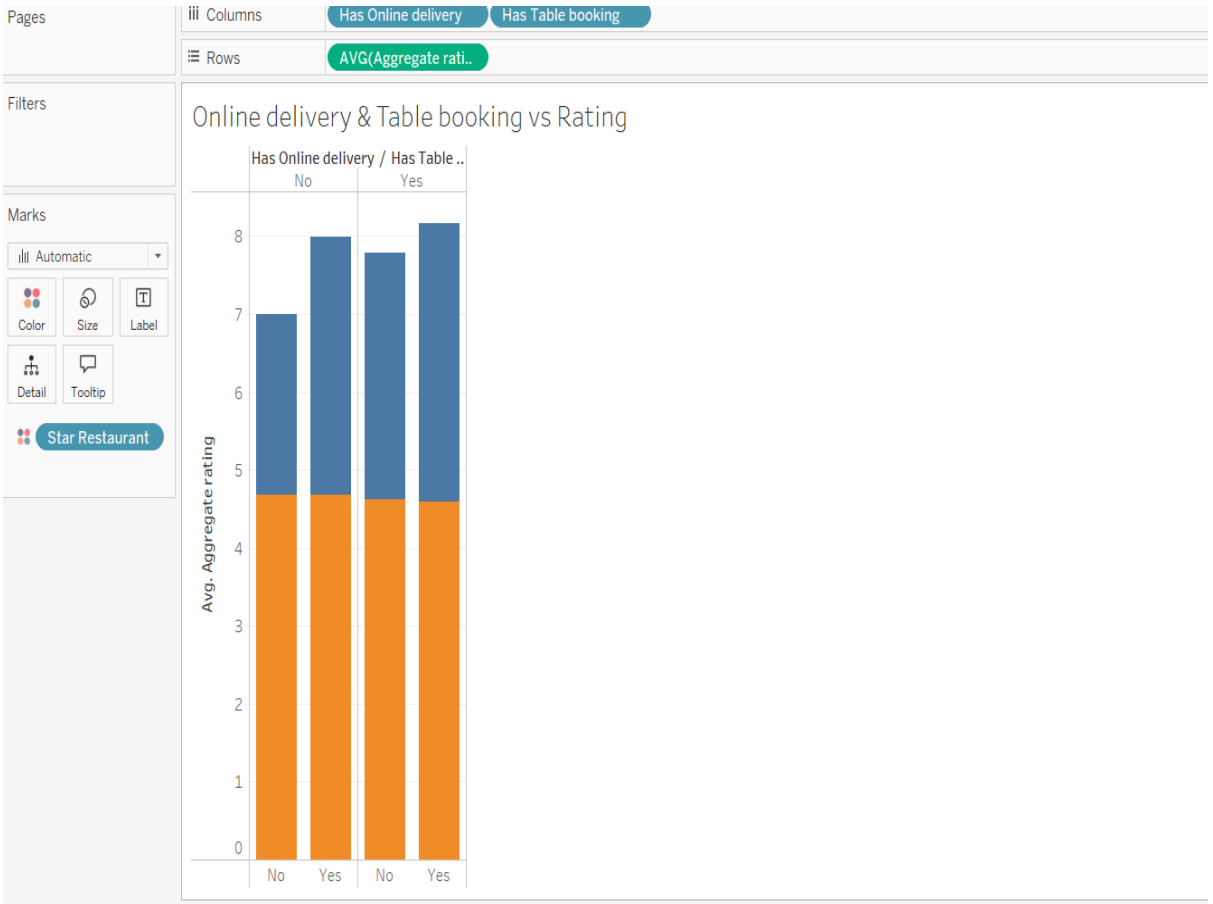
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Star Restaurant
Online delivery & Table booking vs Rating
Has Online delivery / Has Table ..
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No
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